Problem Set 3

Operator Renormalization

1. Consider massive $\lambda\phi^4$ theory, and the composite operator $\hat{O}(x) = \widehat{\varphi^2}(x)$.

Compute the 3-pt function

$$G(q,k,\lambda,m,\Lambda) = \langle \hat{\varphi}(q)\hat{\varphi}(q-k)\hat{O}(k)\rangle$$

to 1-loop with Λ the momentum cut-off.

We can relate this correlator the renormalized one using the eq.

$$G_{\text{amputated}}(k,\lambda,m,\Lambda) = \frac{Z(M)}{Z^{(\varphi^2)}(M)}\, G_{R,\,\text{amput.}}(k,\lambda(M),m^2(M),M)$$

Use this relation to compute the operator renormalization $\gamma^{(\varphi^2)}(\lambda(M))$.

1. Problem 13.2 Peskin-Schroder

2. Problem 13.3 Peskin-Schroder